

AvatarFit

Think Aloud Usability Test — Session Results

1. Overview

Three participants took part in a Think-Aloud usability test of the AvatarFit prototype on 2 June 2026. Sessions were conducted in-person with a single facilitator. Each session lasted approximately 15-20 minutes and followed a structured script covering four tasks: (1) webcam detection, (2) squat exercise, (3) exercise change, and (4) lunge exercise. Participants were asked to verbalize their thoughts throughout each task. Post-task questions were collected immediately after the final task.

Session summary:

Participant	Date	Start time	Session duration
A.A.C	02/06/2026	15:06	~19 min
G.B.P	02/06/2026	15:25	~18 min
L.R.M	02/06/2026	15:43	~17 min

2. Observation Log

Events recorded by the facilitator during each session.

Participant	Time	Task	Observation / Direct Quote	Type
A.A.C	15:06	Warm-up	"I really like it; the avatars are very cute"	Positive
A.A.C	15:07	Launch & detected	Participant very surprised at avatar mirroring her every movement.	Surprise
A.A.C	15:09	Squat	Did not understand at first that the exercise requires a side-on position.	Confusion
A.A.C	15:14	Change exercise	Found the exercise selector immediately and switched without difficulty.	Correct
A.A.C	15:16	Lunge	By this point understood the app logic and found the feedback useful.	Correct
G.B.P	15:25	Warm-up	Finds the app very original; likes the multiple avatar options.	Positive

G.B.P	15:27	Launch & detected	No comment recorded.	None
G.B.P	15:28	Squat	No difficulty; received feedback, corrected posture and improved form.	Positive
G.B.P	15:33	Change exercise	Changed exercise without any problem.	Correct
G.B.P	15:34	Lunge	Did not understand the first correction; after a few tries achieved correct posture.	Correct
L.R.M	15:43	Warm-up	Immediately noticed there were no female avatars.	Surprise
L.R.M	15:44	Launch & detected	No comment recorded.	None
L.R.M	15:45	Squat	Took a couple of seconds to understand the orientation; once started had no problems.	Positive
L.R.M	15:51	Change exercise	Changed exercise without any problem.	Correct
L.R.M	15:52	Lunge	Did not initially realise the exercise requires a sideways position.	Surprise

3. Post-Task Questions

Q1 — What did you find most confusing about the app?

A.A.C	At the beginning, I was a little confused about how the avatar worked and how the corrections worked.
G.B.P	I didn't find anything really confusing; I was fairly sure about what was happening at every moment.
L.R.M	Maybe the most confusing part was realising when a workout was supposed to be done sideways and when facing forward.

Q2 — Was the feedback (red/green skeleton, text tips) clear to you?

A.A.C	Yes, it was very useful because it gave me helpful recommendations and improved my way of doing the exercise.
G.B.P	Yes, it was clear. With the red/green skeleton there is no doubt about whether you are doing it wrong or right.
L.R.M	Yes, completely. I didn't have any doubt about when I was doing the exercise correctly and when I was not.

Q3 — Did you feel the app was correctly judging your exercise form?

A.A.C	Yes, I think it was correct because it was mainly correcting the angles of my body.
G.B.P	Yes. There was one exercise where it took me more than one attempt to understand what I was doing wrong, but once I changed it I could feel the improvement.
L.R.M	Yes, for sure. Most of the time it was asking me to make my movements larger, and that is something I was looking forward to working on.

Q4 — Is there anything you expected the app to do that it did not do?

A.A.C	I was expecting the avatar to give full corrections out loud instead of only telling me that I was doing it incorrectly.
G.B.P	Maybe it would be nice to hear the whole correction out loud.
L.R.M	I think it would be nice to have a female avatar as well.

Q5 — On a scale of 1 to 5, how easy was it to understand the real-time feedback?

A.A.C	4 — Although the recommendations were helpful, most of the time I had to read them rather than hearing them.
G.B.P	4 — All the corrections were clear.
L.R.M	4 — I could understand every piece of feedback I received.

4. Overall Impressions

A.A.C	Considers AvatarFit very useful and original, noting that she normally has nobody to correct her exercise form when working out at home. Highlighted the beautiful design as an additional strength.
G.B.P	Really liked the prototype and found it highly useful. Praised the concept and the intuitive character-selection screen.
L.R.M	Described it as a great implementation she would use regularly for her workouts. Suggested adding a female avatar as the main improvement.

5. Key Findings

OK Feedback clarity validated (all 3 participants, 4/5 score) All three participants rated feedback clarity at 4/5 and confirmed the red/green skeleton left no ambiguity about form correctness. The visual feedback mechanism is working as intended.

OK Design and avatar system well received The character-selection screen and avatar mirroring generated spontaneous positive comments across all sessions, confirming that the gamified visual design is a genuine differentiator.

OK Exercise switching is intuitive All three participants changed from squat to lunge without difficulty, confirming that the exercise-pill navigation is clear and discoverable.

(!) Side-orientation cue is missing (2/3 participants confused) A.A.C (squat) and L.R.M (lunge) were initially confused about needing to stand sideways. There is currently no on-screen instruction guiding users to turn before starting these exercises.

(!) Voice feedback is too minimal (2/3 participants) A.A.C and G.B.P expected to hear the full verbal correction rather than a generic 'wrong form' cue. The speech output needs to describe the corrective action explicitly.

(!) No female avatar option (1 participant) L.R.M noted immediately that all avatar characters are male-presenting. This is a representational gap that affects perceived inclusivity.

6. Design Recommendations

R1 — Add side-orientation instruction	Display a visible cue (animated arrow or text banner) prompting the user to turn sideways before starting the squat or lunge. This addresses the most common confusion observed across sessions.
R2 — Expand verbal feedback to full corrections	Update speech synthesis to describe the corrective action explicitly (e.g., 'Lower your hips more' or 'Keep your back straight') rather than simply signalling incorrect form. Requested by 2/3 participants.
R3 — Add a female avatar option	Introduce at least one female-presenting character in the avatar selection screen to improve inclusivity.
R4 — Consider an onboarding step	A brief tutorial or first-use overlay explaining key interface elements (skeleton feedback, exercise switching, voice commands) could reduce initial confusion for new users.

7. Conclusions

The Think-Aloud pilot test provides early but consistent usability evidence for AvatarFit. The core concept was understood and appreciated by all three users without prior instruction. An average perceived feedback ease score of 4/5 confirms that the visual feedback channel is working effectively.

The two most actionable issues are the absence of a side-orientation cue for squat and lunge, and the limited verbosity of the voice feedback. The request for a female avatar is a medium-priority design improvement.

Overall, the evaluation validates the primary HCI problem statement from the Week 1 survey: users working out at home lack real-time form correction. AvatarFit directly addresses this need, and participant reactions indicate a high likelihood of regular adoption once the noted issues are resolved.